

# Powerball Rolling Window Overlap Analysis Report

## Project Overview

This is a comprehensive statistical analysis study of official New York State Powerball lottery data. By analyzing 1,967 lottery drawings (2010-2026), we explore the overlap patterns of white ball numbers across consecutive draws. The core research method uses rolling window analysis, examining windows of 2-10 consecutive draws and constructing a complete empirical probability matrix.

## Dataset

Source: Official New York State Powerball Lottery Records Time Period: January 2010 - July 2026 Sample Size: 1,967 drawings Data Quality: Verified duplicate-free, fully cleaned and standardized

## Methodology

For each window size (2 to 10 consecutive drawings): 1. Extract all possible consecutive N-drawing windows from the dataset 2. For each window, count how many white ball numbers appear 2+ times 3. Collect all overlap counts 4. Calculate mean, median, min, max, and standard deviation 5. Build probability distribution 6. Assemble final probability matrix

## Key Findings

Key Observations: • Larger windows show more overlaps (consistent with probability theory) • Window Size 2: Average 0.38 overlapping numbers • Window Size 5: Average 3.30 overlapping numbers • Window Size 10: Average 11.39 overlapping numbers • Distribution shifts with window size • Small windows: concentrated on low overlaps • Large windows: broader distribution, skewed toward high overlaps

## Probability Matrix

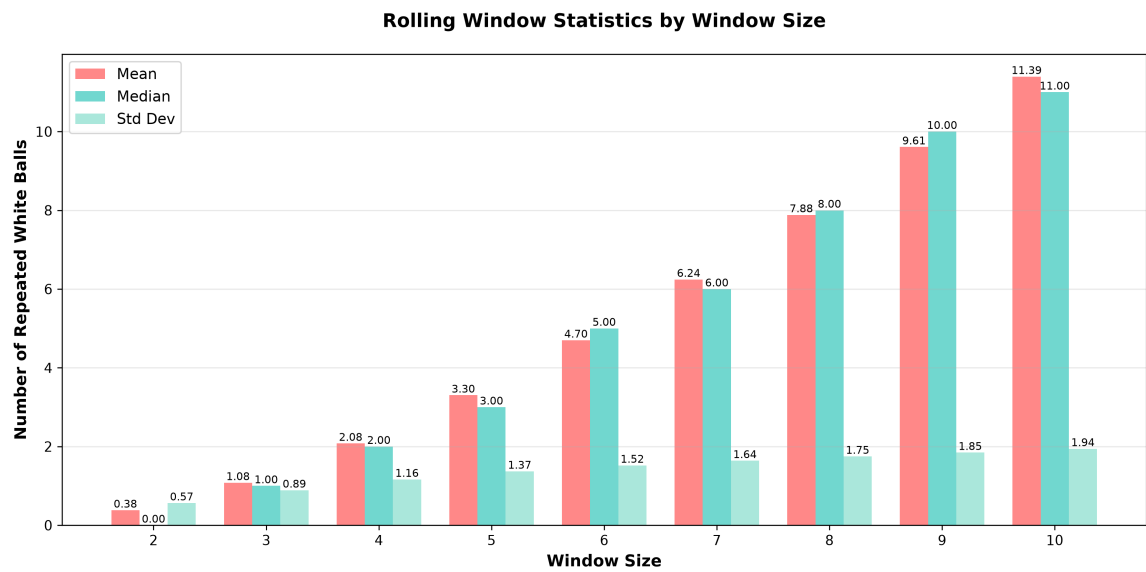
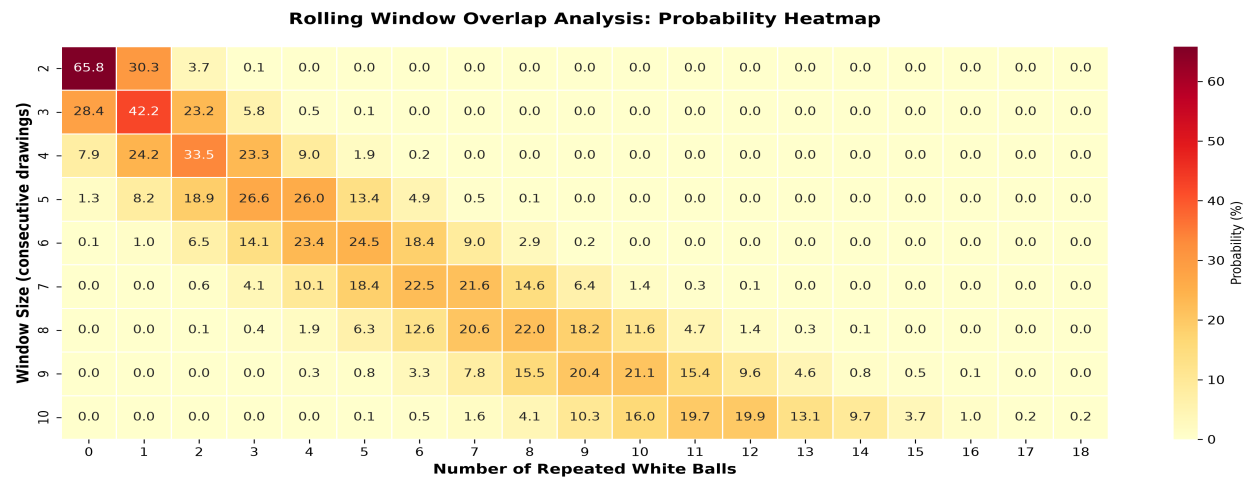
| Window_Size | Repeated_0 | Repeated_1 | Repeated_2 | Repeated_3 | Repeated_4 | Repeated_5 | Repeated_6 | Repeated_7 | Repeated_8 |
|-------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 2.0         | 65.82      | 30.32      | 3.71       | 0.15       | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        |
| 3.0         | 28.35      | 42.19      | 23.16      | 5.75       | 0.51       | 0.05       | 0.0        | 0.0        | 0.0        |
| 4.0         | 7.94       | 24.19      | 33.5       | 23.27      | 8.96       | 1.88       | 0.25       | 0.0        | 0.0        |
| 5.0         | 1.32       | 8.25       | 18.95      | 26.64      | 25.98      | 13.4       | 4.89       | 0.51       | 0.05       |
| 6.0         | 0.1        | 1.02       | 6.52       | 14.07      | 23.39      | 24.52      | 18.35      | 8.97       | 2.85       |
| 7.0         | 0.0        | 0.0        | 0.61       | 4.08       | 10.05      | 18.36      | 22.54      | 21.62      | 14.58      |
| 8.0         | 0.0        | 0.0        | 0.05       | 0.36       | 1.94       | 6.28       | 12.55      | 20.61      | 22.04      |
| 9.0         | 0.0        | 0.0        | 0.0        | 0.0        | 0.31       | 0.77       | 3.27       | 7.76       | 15.52      |
| 10.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.05       | 0.46       | 1.58       | 4.14       |

*Note: Table shows first 8 overlap columns. Full matrix in rolling\_window\_matrix.csv*

## Entertainment Prediction Analysis

Entertainment Analysis: Avoiding "Hot Numbers" Question: If a player avoids the "most frequently drawn numbers," does it improve odds? Analysis Results: • Number frequency differences are minimal (hottest 175x vs coldest 88x) • This variation is completely consistent with random fluctuation • Avoiding "hot" or "cold" numbers does NOT change winning probability Why Not? • Each lottery drawing is independent • Past frequency doesn't affect future outcomes • Mathematically, ANY selection of 5 numbers has identical probability Entertainment Recommendations: ✓ Choose numbers you like - odds are the same ✓ Don't change your choice based on history ✓ Treat lottery as entertainment, not investment

Visualizations



## Interpretation & Insights

This study provides a complete statistical description of white ball overlap patterns in historical Powerball data. Important Notes: • This is historical analysis, NOT prediction • Past patterns cannot predict future outcomes • Lottery outcomes are inherently random and unpredictable • No strategy exists that can improve odds

## Limitations

• Time-Specific: Analysis based on 2010-2026 data only • Mechanism Changes: Lottery rules may change in future • No Predictive Value: Cannot be used to predict future drawings • Sample Size: 1,967 draws are substantial but still small in full probability space